

TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

Advanced cellular IoT module with enhanced connectivity

Cavli Wireless has launched CQS290, a cellular Android module from the Cavli C-Series. The long-term evolution (LTE) Cat 4 module offers data rates up to 150 Mbit/s for downloads and 50



Mbit/s for uploads and includes 2G fallback. It consists of a Cortex A53 quad-core CPU and an Adreno 702 GPU. The module is available in two regional variants

and comes with the Qualcomm Location Suite Gen 9VT engine, supporting GNSS systems like GPS and GLONASS. It is suitable for various applications, including two-wheelers, telematics, fleet management, and point-of-sale terminals. The module also offers an optional integrated eSIM. The Hubble Stack Intelligence adds to the CQS290's functions, enabling remote diagnostics, monitoring, and debugging through Hubble Lens, a Cavli Hubble IoT platform feature. This reduces the need for physical intervention and aids in IoT scalability.

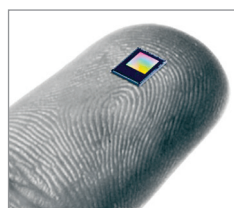
Cavli Wireless

<https://www.cavliwireless.com>

Event-based vision sensor for energy-efficient Edge AI

Prophesee SA has launched the GenX320 Event-based Metavision sensor. This is the first event-based vision sensor for ultra-low-power Edge AI vision devices. As the fifth generation of Metavision technology, it has

a 3x4mm die size and extends its technology to various intelligent Edge market segments. These include AR/VR headsets, security and monitoring systems, touchless displays, eye-tracking features, always-on smart IoT devices, and more. The GenX320 offers capabilities like speed, low latency, dynamic range, power efficiency, and privacy in various applications.



This sensor is designed to meet the requirements of integrating event sensing in embedded vision systems with energy, computing power, and size constraints. The sensor provides vision capabilities with low power consumption in various operating and lighting conditions.

Prophesee SA

<https://www.prophesee.ai>

High-performance microcontrollers with AI features

Renesas Electronics Corporation has launched microcontrollers (MCUs) that provide over 3000 CoreMark points, offering deterministic, low latency, real-time operation for various applications. The RA8 series MCUs are the industry's



first to use the Arm Cortex-M85 processor, achieving a 6.39 CoreMark/MHz1 performance. This performance allows system

designers to use the RA MCUs in applications that typically require microprocessors (MPUs). The MCUs incorporate Arm Helium technology, an M-Profile

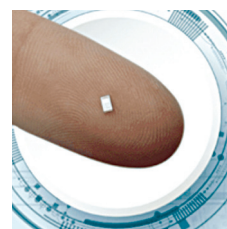
vector extension from Arm, which provides up to four times the performance for digital signal processing (DSP) and machine learning (ML) compared to MCUs with the Arm Cortex-M7 processor. The RA8 series MCUs support edge and endpoint devices in implementing natural language processing for voice AI and predictive maintenance applications by accelerating neural network processing with Helium.

Renesas Electronics Corporation

<https://www.renesas.com>

First piezoelectric sensor for automotive interfaces

UltraSense TouchPoint Q from UltraSense Systems is the first piezoelectric strain sensor developed to enhance the touch experience in automotive interfaces. It offers calibratable force thresholds for instant robustness and is immune to



temperature sensitivity, mechanical stresses, and ageing.

Additionally, it remains stable against signal drift caused by

adhesive thermal expansion mismatch, a common issue in other force-sensing technologies. The sensor's unique micro-electro-mechanical system (MEMS)-based QuadForce architecture incorporates four strain sensors in each component. This enables differential sensing and pattern recognition capabilities, especially when integrated with machine learning. The sensor is also fully qualified to AEC Q100 Grade 2 standards, ensuring reliability in temperatures ranging from -40 to +105°C.

UltraSense Systems

<https://ultrasensesys.com>